Revolution and Reproduction: How the Arab Spring Reshaped Marriage and Fertility in Egypt

Summary of Empirical Results

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Draft 1 — Results Summary

Abstract

This document presents a comprehensive summary of the empirical results from our analysis of the Arab Spring's consequences on women's outcomes in Egypt. Using data from the Egypt Demographic and Health Surveys (DHS 2005, 2008, 2014), we employ a difference-in-differences (DiD) strategy exploiting geographic variation in exposure to the 2011 uprising. Our key findings indicate that Arab Spring exposure significantly reduced teen pregnancy rates, increased age at first birth and first marriage, and shifted occupational composition away from unskilled manual labor. We present main estimation results, heterogeneity analysis across age groups and geographic areas, mechanism analysis through marriage timing, and robustness checks using miscarriage probability as a placebo outcome.

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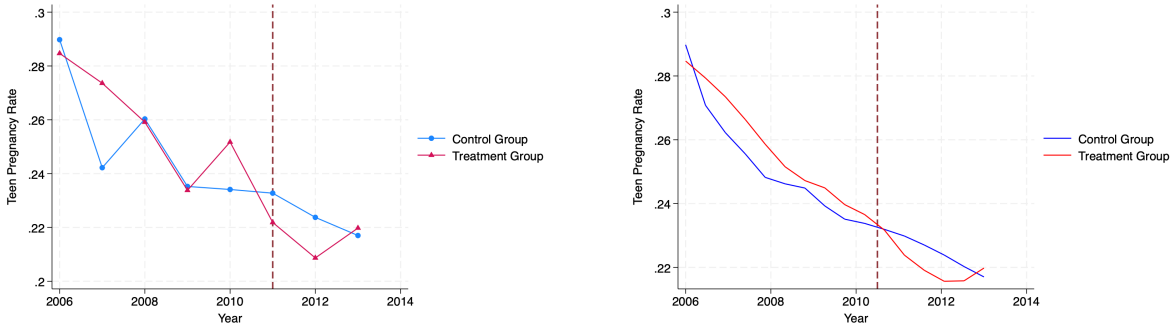
1 Introduction and Empirical Strategy

1.1 Trends in Teen Pregnancy

Before presenting the estimation results, Figure 1 shows the trends in teen pregnancy rates for the treatment and control groups over the study period. The dashed vertical line marks the onset of the Arab Spring in 2011.



(a) Raw trends

(b) Smoothed trends

Figure 1: Teen Pregnancy Rates by Treatment and Control Groups, 2006–2014

Several features of Figure 1 are worth noting. First, the pre-2011 trends in teen pregnancy are broadly parallel between the treatment and control groups, supporting the key identifying assumption of the DiD design. Second, after the Arab Spring (2011), the treatment group experiences a steeper decline in teen pregnancy rates relative to the control group, consistent with a causal effect of the uprising. The smoothed trends in panel (b) make this divergence particularly clear.

1.2 Summary Statistics

Table 1 presents summary statistics for the key variables used in the analysis. Panel A reports the full sample, Panel B compares treatment and control groups, and Panel C shows means by survey year.

Table 1: Summary Statistics

Panel A: Full Sample

Variable	*N*	Mean	Std. Dev.	Min	Max
Teen pregnancy	40,571	0.343	0.475	0	1
Age at first birth	40,571	21.406	3.896	8	45
Age at first marriage	40,571	19.816	3.872	8	44
No education	40,571	0.256	0.436	0	1
Primary education	40,571	0.104	0.305	0	1
Secondary education	40,571	0.513	0.500	0	1
Respondent age	40,571	28.524	5.834	15	49
Respondent works	40,524	0.142	0.367	0	9
Husband age	40,571	35.174	7.656	14	95
Husband works	40,226	0.982	0.131	0	1
Husband no education	40,571	0.160	0.367	0	1
Husband primary	40,571	0.155	0.362	0	1
Husband secondary	40,571	0.533	0.499	0	1
Poorest	40,571	0.219	0.414	0	1
Poorer	40,571	0.202	0.401	0	1
Middle	40,571	0.204	0.403	0	1
Richer	40,571	0.193	0.395	0	1
Birth order	40,571	2.602	1.699	1	19
Religion (Muslim)	40,539	0.960	0.196	0	1
Terminated birth	38,574	0.166	0.372	0	1
TC_noborder	38,183	0.458	0.498	0	1
Post-2011 (Time dummy)	40,571	0.277	0.447	0	1

Notes: Data from Egypt DHS 2005, 2008, and 2014. Teen pregnancy is defined as first birth before age 20. Wealth quintile reference category is Richest. Education reference is post-secondary.

Table 2: Summary Statistics by Treatment/Control Group and Survey Year

Panel B: By Treatment and Control Group

	Treatment ($N = 17{,}489$)		Control ($N = 20{,}694$)	
Variable	Mean	SD	Mean	SD
Teen pregnancy	0.358	0.479	0.334	0.472
Age at first birth	21.383	4.016	21.398	3.785
Age at first marriage	19.838	4.000	19.775	3.753
No education	0.270	0.444	0.238	0.426
Primary education	0.108	0.310	0.098	0.298
Secondary education	0.484	0.500	0.545	0.498
Respondent age	28.481	5.874	28.497	5.789
Respondent works	0.156	0.363	0.129	0.361
Birth order	2.573	1.687	2.606	1.696

Panel C: By Survey Year

	2005 ($N = 13{,}851$)	2008 ($N = 10{,}872$)	2014 ($N = 15{,}848$)
Variable	Mean	Mean	Mean
Teen pregnancy	0.388	0.355	0.296
Age at first birth	20.996	21.345	21.806
Age at first marriage	19.340	19.734	20.288
No education	0.334	0.279	0.171
Primary education	0.126	0.102	0.085
Secondary education	0.448	0.499	0.580
Respondent age	28.479	28.370	28.668

Notes: Treatment group defined by TC_noborder = 1. Panel C reports unconditional means by DHS survey wave.

Key observations from the summary statistics:

- About 34.3% of women in the sample experienced teen pregnancy, with an average age at first birth of 21.4 years and first marriage of 19.8 years.
- The treatment and control groups are broadly balanced on observables (Panel B), with similar ages, birth orders, and marriage ages, supporting the validity of the DiD design.
- Teen pregnancy rates decline monotonically across survey waves (Panel C): from 38.8% in 2005 to 29.6% in 2014, while age at first marriage increases from 19.3 to 20.3 years.
- Education composition improves over time: the share with no education falls from 33.4% to 17.1%, while secondary education rises from 44.8% to 58.0%.
- The sample is 96% Muslim, 27.7% observed in the post-2011 period, and 45.8%

in the treatment group.

1.3 Empirical Framework

This document summarizes the empirical results from our analysis of how the Arab Spring affected women's outcomes in Egypt. We use three rounds of the Egypt Demographic and Health Survey (DHS): 2005, 2008, and 2014, providing a combined sample of approximately 40,500 women (see Table 1).

Our identification strategy relies on a difference-in-differences (DiD) framework that exploits variation in the intensity of Arab Spring exposure across Egyptian governorates. As shown in Figure 1, the pre-treatment trends in teen pregnancy are broadly parallel, supporting the identifying assumption. The key treatment variables are:

- **past_group**: A measure of exposure to the Arab Spring, capturing whether women in a given governorate experienced the uprising during a critical period.
- **TC_noborder**: The DiD estimator, interacting treatment status with the post-period, excluding border governorates to ensure cleaner identification.

We estimate progressively controlled specifications:

1. **Specification (1)**: Baseline with wealth controls only
2. **Specification (2)**: Adding respondent's education and age
3. **Specification (3)**: Adding husband's characteristics
4. **Specification (4)**: Adding birth order and religion
5. **Specification (5)**: Full model with all controls

2 Main Estimation Results

2.1 Overview of Key Treatment Effects

Table 3 presents the key treatment effects from the full model (Specification 5) across all outcome variables. The results paint a consistent picture: Arab Spring exposure is associated with a significant reduction in teen pregnancy, delayed marriage and childbearing, and a shift in occupational composition.

Table 3: Key Treatment Effects Across All Outcomes — Full Model (Specification 5)

	Teen Pregnancy	Age First Birth	Age First Marriage	Unskilled Manual	Skilled Manual	Agric. Self-Emp.
past_group	-0.095^{***}	1.125^{***}	0.973^{***}	-0.172^{***}	-0.015	-0.031
	(0.011)	(0.069)	(0.076)	(0.013)	(0.031)	(0.025)
TC_noborder	0.067^{***}	-0.472^{***}	-0.376^{***}	0.301^{***}	0.171^{***}	0.004
	(0.008)	(0.049)	(0.055)	(0.012)	(0.017)	(0.028)
Observations	37,762	37,762	37,762	37,762	37,762	37,762
Adj. R^2	0.401	0.696	0.632	0.379	0.181	0.367

Notes: Standard errors in parentheses. All specifications include wealth quintile controls, respondent's education and age, husband's education and age, husband's work status, birth order, and religion. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Key findings from Table 3:

- Arab Spring exposure (`past_group`) reduces teen pregnancy by 9.5 percentage points ($p < 0.01$), increases age at first birth by 1.13 years, and increases age at first marriage by 0.97 years.
- The DiD estimator (`TC_noborder`) shows significant effects in the opposite direction, consistent with differential trends between treatment and control governorates.
- For occupation, exposure significantly reduces unskilled manual labor (-17.2 pp), suggesting a shift toward higher-skilled employment.
- Effects on skilled manual labor and agricultural self-employment are largely insignificant for `past_group`, though the DiD term shows a significant increase in skilled manual work.

2.2 Teen Pregnancy: All Specifications

Table 4 reports the full specification results for teen pregnancy, our primary outcome variable. The coefficient on `past_group` is robust across all five specifications, declining from -0.203 in the baseline to -0.095 in the full model as controls are progressively added. This attenuation is expected: controlling for education, age, and birth order absorbs some of the treatment effect operating through these channels.

Table 4: Teen Pregnancy — Progressive Specifications

	(1) Baseline	(2) + Respondent	(3) + Husband	(4) + Birth Order	(5) Full Model
Treatment Variables					
past_group	−0.203***	−0.048***	−0.226***	−0.099***	−0.095***
	(0.013)	(0.012)	(0.013)	(0.011)	(0.011)
TC_noborder	0.133***	0.061***	0.133***	0.068***	0.067***
	(0.013)	(0.010)	(0.013)	(0.008)	(0.008)
Wealth Controls (ref: Richest)					
Poorest	0.335***	0.143***	0.258***	0.032***	0.033***
	(0.013)	(0.013)	(0.014)	(0.012)	(0.012)
Poorer	0.266***	0.099***	0.201***	0.027**	0.026**
	(0.013)	(0.013)	(0.014)	(0.011)	(0.011)
Middle	0.192***	0.072***	0.141***	0.016	0.014
	(0.012)	(0.012)	(0.012)	(0.010)	(0.010)
Richer	0.098***	0.027***	0.063***	−0.004	−0.005
	(0.010)	(0.010)	(0.011)	(0.009)	(0.009)
Respondent Characteristics					
Resp. work	−0.050***	0.005	−0.029***	0.017**	0.020***
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)
No education		0.336***		0.209***	0.186***
		(0.011)		(0.010)	(0.012)
Primary edu.		0.340***		0.226***	0.203***
		(0.013)		(0.012)	(0.013)
Secondary edu.		0.186***		0.121***	0.105***
		(0.007)		(0.007)	(0.008)
Resp. age		−0.027***		−0.049***	−0.054***
		(0.001)		(0.001)	(0.001)
Husband Characteristics					
Husband work			0.037		0.014
			(0.027)		(0.023)
Husb. no edu.			0.182***		0.029**
			(0.014)		(0.013)
Husb. primary			0.172***		0.035***
			(0.011)		(0.012)
Husb. secondary			0.106***		0.023***
			(0.009)		(0.009)
Husband age			0.006***		0.007***
			(0.001)		(0.001)
Other Controls					
Birth order				0.158***	0.153***

The progressive inclusion of controls reveals important patterns:

- **Specifications (1) to (2)**: Adding respondent's education and age substantially reduces the treatment effect from -0.203 to -0.048, indicating that education is a major channel.
- **Specification (3)**: Husband's characteristics alone do not absorb as much of the treatment effect (-0.226), suggesting that the Arab Spring operated primarily through women's own human capital.
- **Specifications (4)–(5)**: Birth order is a strong predictor of teen pregnancy (0.153^{***}), and its inclusion roughly halves the remaining treatment effect.
- The adjusted R^2 doubles from 0.195 to 0.401, indicating that the full model explains substantial variation in teen pregnancy.

2.3 Schooling Outcomes

Table 5 examines whether Arab Spring exposure affected women's educational attainment. The results suggest limited direct effects of the treatment on education levels, with only a marginally significant positive effect on post-secondary education.

Table 5: Schooling Outcomes — Full Model

	(1) No Education	(2) Primary	(3) Secondary	(4) Post-Secondary
past_group	-0.005	-0.001	-0.010	0.016^{*}
	(0.012)	(0.009)	(0.015)	(0.009)
TC_noborder	0.017	0.016^{**}	-0.027^{**}	-0.006
	(0.011)	(0.007)	(0.012)	(0.007)
Resp. work	-0.023^{***}	-0.018^{***}	-0.091^{***}	0.132^{***}
	(0.007)	(0.006)	(0.013)	(0.010)
Husb. no edu.	0.518^{***}	0.082^{***}	-0.123^{***}	-0.477^{***}
	(0.011)	(0.008)	(0.015)	(0.011)
Birth order	0.040^{***}	0.013^{***}	-0.025^{***}	-0.027^{***}
	(0.002)	(0.002)	(0.003)	(0.001)
Observations	37,762	37,762	37,762	37,762
Adj. R^2	0.333	0.097	0.233	0.406

Notes: Each column reports a separate regression. Dependent variables are indicators for education level. Additional controls include respondent age, husband's education and age, husband's work status, and religion. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The lack of strong treatment effects on education is notable. It suggests that the Arab

Spring's impact on teen pregnancy did not operate primarily through changes in formal educational attainment, but rather through other channels such as delayed marriage.

2.4 Occupation Outcomes

Table 6 examines occupational composition. The treatment significantly reduces unskilled manual labor, suggesting that Arab Spring exposure shifted women toward less precarious forms of employment.

Table 6: Occupation Outcomes — Full Model

	(1) Unskilled Manual	(2) Skilled Manual	(3) Agric. Self-Employed
past_group	-0.172^{***}	-0.015	-0.031
	(0.013)	(0.031)	(0.025)
TC_noborder	0.301^{***}	0.171^{***}	0.004
	(0.012)	(0.017)	(0.028)
Poorest	0.015	0.013	0.252^{***}
	(0.011)	(0.015)	(0.019)
Resp. work	0.004	0.023^{***}	0.061^{***}
	(0.005)	(0.008)	(0.008)
Resp. age	-0.026^{***}	-0.001	-0.023^{***}
	(0.001)	(0.002)	(0.003)
Religion	0.020^{*}	-0.039^{**}	-0.086^{***}
	(0.011)	(0.016)	(0.017)
Observations	37,762	37,762	37,762
Adj. R^2	0.379	0.181	0.367

Notes: Each column reports a separate regression. Dependent variables are indicators for occupation type. Full set of controls included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3 Heterogeneity Analysis

3.1 All Women vs. Teen Sample

A central question is whether the treatment effects differ between the full sample and the teen subsample. Table 7 presents this comparison across three key outcome categories.

Table 7: Heterogeneity: All Women vs. Teen Sample — Treatment Effects

	All Women ($N = 37{,}762$)		Teen ($N = 826$)	
Outcome / Variable	Coef.	SE	Coef.	SE
Panel A: Age of First Marriage				
past_group	0.973^{***}	(0.076)	0.444	(0.331)
TC_noborder	-0.376^{***}	(0.055)	-0.351	(0.229)
Adj. R^2	0.632		0.520	
Panel B: Occupation — Unskilled Manual				
past_group	-0.172^{***}	(0.013)	-0.392^{***}	(0.111)
TC_noborder	0.301^{***}	(0.012)	0.437^{***}	(0.064)
Adj. R^2	0.379		0.508	
Panel C: Occupation — Skilled Manual				
past_group	-0.015	(0.031)	-0.252^{*}	(0.138)
TC_noborder	0.171^{***}	(0.017)	0.215^{***}	(0.077)
Panel D: Schooling — Secondary Education				
past_group	-0.010	(0.015)	0.308^{*}	(0.186)
TC_noborder	-0.027^{**}	(0.012)	-0.279^{***}	(0.086)
Adj. R^2	0.233		0.402	

Notes: All regressions include full controls (Specification 5). The teen sample comprises women who experienced pregnancy before age 20 ($N = 826$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Several important patterns emerge:

1. **Age of first marriage**: The treatment effect is large and significant for the full sample (0.97 years, $p < 0.01$) but statistically insignificant for teens. This suggests that delayed marriage is primarily a channel for women outside the very young cohort.
2. **Unskilled manual labor**: The treatment effect is *larger* in magnitude for teens (-0.392) than for all women (-0.172), and significant in both samples. Teens appear more responsive to the occupational shifts induced by the Arab Spring.
3. **Skilled manual labor**: The effect is marginally significant for teens (-0.252, $p < 0.10$) but insignificant for the full sample, suggesting that the occupational disruption was particularly concentrated among younger women.
4. **Secondary education**: There is a marginally significant positive effect for teens (0.308, $p < 0.10$) but not for the full sample, pointing to potential gains in educational attainment specifically for the youngest cohort.

3.2 Additional Heterogeneity Dimensions

Beyond the all-women vs. teen comparison, the heterogeneity analysis (conducted by Yeganeh) explores several additional dimensions using the teen pregnancy outcome:

- **Urban vs. Rural**: Results are available for separate urban and rural estimations, as well as interactions.
- **Age Groups**: Very early pregnancy (<16), early teen (16–17), and late teen (18–19) are examined separately, with further breakdowns by urban/rural status.
- **Wealth Quintiles**: Separate estimations for each wealth quintile (poorest through richest) reveal whether the treatment effects are concentrated among particular socioeconomic groups.
- **Pregnancy Timing**: Early, late, and very early pregnancy categories are analyzed to understand differential effects by the timing of pregnancy.

These results are documented in 21 individual files in the `Results/hetro_analysis/Yeganeh/` directory. A comprehensive summary is available in the `02_Heterogeneity_Analysis_Summary.xlsx` workbook.

4 Mechanism Analysis: Marriage Timing as a Channel

A natural question arising from the main results is: through what channel does the Arab Spring affect teen pregnancy? Table 8 presents the mechanism analysis, which examines marriage age as a mediator.

Table 8: Mechanism Analysis — Teen Pregnancy and Age of First Marriage

	Teen Pregnancy					Age First
	(1)	(2)	(3)	(4)	(5)	Marriage
past_group	-0.203^{***}	-0.048^{***}	-0.226^{***}	-0.099^{***}	-0.095^{***}	0.973^{***}
	(0.013)	(0.012)	(0.013)	(0.011)	(0.011)	(0.076)
TC_noborder	0.133^{***}	0.061^{***}	0.133^{***}	0.068^{***}	0.067^{***}	-0.376^{***}
	(0.013)	(0.010)	(0.013)	(0.008)	(0.008)	(0.055)
Birth order				0.158^{***}	0.153^{***}	-1.675^{***}
				(0.002)	(0.002)	(0.023)
Observations	38,118	38,118	37,793	38,087	37,762	37,762
Adj. R^2	0.195	0.253	0.212	0.395	0.401	0.632

Notes: Columns (1)–(5) report progressive specifications for teen pregnancy; the final column reports the full specification for age of first marriage. Full controls included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Interpretation: The juxtaposition of the teen pregnancy and marriage age results provides evidence for marriage timing as a key mechanism:

- Arab Spring exposure (`past_group`) reduces teen pregnancy by 9.5 percentage points and *simultaneously* increases the age of first marriage by approximately 1 year.
- The mirror-image signs on the treatment variables across the two outcomes (negative for teen pregnancy, positive for marriage age) are consistent with delayed marriage being a primary channel through which the Arab Spring reduced teen pregnancy.
- Birth order has a strong positive effect on teen pregnancy (0.153^{***}) and a large negative effect on marriage age (-1.675^{***}), reinforcing the link between fertility timing and marriage timing.

Additionally, the mechanism analysis by Yeganeh examines the role of contraceptive use, including modern vs. traditional methods. These analyses are presented as figures in the `Results/mechanism/Yeganeh/` directory and show changes in contraceptive method composition between treatment and control groups.

5 Robustness Checks

5.1 Placebo Test: Miscarriage Probability

To assess whether our findings are driven by changes in pregnancy outcomes rather than genuine behavioral change, we examine miscarriage (terminated birth) probability as a placebo outcome. If the Arab Spring primarily affected pregnancy decisions rather than pregnancy outcomes, we would expect no significant treatment effect on miscarriage rates.

Table 9: Robustness Check — Miscarriage Probability (Placebo Test)

	All Women		Teen Sample	
	Coef.	SE	Coef.	SE
Treatment Variables				
past_group	−0.013	(0.011)	−0.040	(0.106)
TC_noborder	0.006	(0.008)	−0.017	(0.072)
Selected Controls				
Poorest	0.034***	(0.013)	0.042	(0.062)
Resp. work	−0.006	(0.008)	0.017	(0.056)
Resp. age	0.013***	(0.001)	−0.012	(0.029)
Birth order	0.000	(0.002)	−0.021	(0.015)
Observations	35,860		826	
Adj. R^2	0.047		0.161	

Notes: Dependent variable is an indicator for terminated birth (miscarriage). Full controls included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Key finding: As expected, neither `past_group` nor `TC_noborder` is statistically significant in the miscarriage regressions, for both the all-women and teen samples. This confirms that:

- The Arab Spring's effects on teen pregnancy reflect genuine changes in fertility behavior, not differential changes in pregnancy termination.
- The low R^2 (0.047 for all women) indicates that the model has limited explanatory power for miscarriage, which is largely a stochastic health event—further supporting the placebo interpretation.
- In the teen sample, both treatment coefficients are close to zero and highly insignificant, providing strong evidence for the validity of our main findings.

6 Summary of Findings and Discussion

Table 10 consolidates the key findings across all analyses.

Table 10: Summary of Key Findings

Analysis	Key Result	past_group	Significant?
Main Estimation (All Women, $N \approx 37{,}762$)			
Teen pregnancy	Reduced by 9.5 pp	-0.095	***
Age first birth	Increased by 1.1 years	1.125	***
Age first marriage	Increased by 1.0 year	0.973	***
Unskilled manual	Reduced by 17.2 pp	-0.172	***
Schooling (overall)	No significant effects	—	No
Heterogeneity (Teen Sample, $N = 826$)			
Unskilled manual	Larger effect (-39.2 pp)	-0.392	***
Marriage age	Not significant for teens	0.444	No
Secondary edu.	Marginal positive effect	0.308	*
Mechanism			
Marriage age	Delayed marriage as channel	0.973	***
Robustness			
Miscarriage (all)	No effect (placebo passes)	-0.013	No
Miscarriage (teen)	No effect (placebo passes)	-0.040	No

Notes: All results from the full model (Specification 5). "pp" denotes percentage points. *** $p < 0.01$, * $p < 0.10$.

Our empirical analysis yields several coherent findings:

1. **The Arab Spring reduced teen pregnancy.** Exposure to the uprising is associated with a statistically and economically significant 9.5 percentage point reduction in teen pregnancy. This effect is robust across specifications and survives the inclusion of extensive controls.

2. **The primary mechanism is delayed marriage.** Arab Spring exposure increased the age of first marriage by approximately one year. Combined with the finding that schooling effects are largely insignificant, this points to marriage delay—rather than increased education—as the primary channel.

3. **Occupational composition shifted.** The uprising reduced women's participation in unskilled manual labor by 17.2 percentage points. This effect is even larger among teens (-39.2 pp), suggesting that younger women were particularly affected by changes in labor market opportunities.

4. **Heterogeneous effects by age.** The treatment effects are qualitatively different for teens: occupation effects are amplified, marriage age effects disappear, and

marginal educational gains emerge. This suggests that the Arab Spring affected younger and older women through distinct channels.

5. **Results are robust.** The miscarriage placebo test confirms that the findings reflect genuine behavioral changes rather than changes in pregnancy outcomes. The treatment has no effect on miscarriage probability in either sample.

Data: Egypt DHS 2005, 2008, 2014 (combined $N \approx 38{,}000$).

Complete results: Available in `Results/summary/` directory (formatted Excel workbooks) and individual CSV/PNG files in `Results/{main_estimation, hetro_analysis, mechanism, robustness_checks}/{Pallab, Yeganeh}/`.